SECTION 14

External Interfaces

Main window ribbon: **Interfaces** tab

External interfaces allow data transfer between CAESAR II and other software applications. In most cases, data transfer is *from* a drawing or analysis package *to* CAESAR II. However:

- Most of the interfaces are CAD interfaces. The exceptions are LIQT, AFT IMPULSE™, PIPENET™, Pipeplus, FlowMaster®, the CAESAR II Data Matrix, and the CAESAR II Neutral File.
- CAD interfaces (CADPIPE, Intergraph Smart 3D PCF, Intergraph PDS, and PCF) are intended to transfer piping geometry into CAESAR II. The resulting CAESAR II input must be thoroughly checked, with loads, restraints, and other specifics added.
- CAESAR II Neutral File transfers both to and from CAESAR II.
- Intergraph CADWorx Plant is the Intergraph CAS piping design and drafting software for the AutoCAD environment. It provides a seamless, bi-directional interface between CADWorx and CAESAR II without a translation procedure. Data may be completely and seamlessly transferred between CAESAR II and Intergraph CADWorx Plant, without creating any neutral files or going through any intermediate steps.
- The CAESAR II Neutral File and the Intergraph CADWorx Plant interfaces are capable of transferring 100% of the data that comprises the _A (input) file.
- LIQT, AFT IMPULSE, PIPENET, PipePlus, and FlowMaster are transient analysis packages for liquids in piping networks that calculate pressure imbalances as a function of time. The CAESAR II interface converts this LIQT output from these packages to create force response spectra for CAESAR II dynamic input.
- The interfaces typically prompt you for a file name, transfer the data, and then prompt for another file name. This cycle continues either until a blank file name is encountered or you cancel the process.
- Before beginning an interface to CAESAR II, follow the requirements of the CAESAR II Neutral File interface. This enables all of the spreadsheet data to be transferred.

CAESAR II Neutral File

The CAESAR II neutral file, an ASCII text file with the .cii extension, enables access to any particular data item from an _A input file, to enable a complete _A file to be built from a CAD application. The .ciil file can be used to send data either to or from the _A file. The name of the file used or generated by this interface is the CAESAR II jobname.

NOTE The content and format described in this section is subject to change as a function of the enhancements made to CAESAR II. Every effort is being made to keep "drastic" changes to a minimum.

Several third-party CAD applications, such as AVEVA®'s PDMS®' and Jacobus' PlantSpace™, also support this neutral file. If you prefer, instead of launching this interface from **Tools** >

External Interface, the processor can be run in "batch mode" from either a batch file or the command line as shown below.

```
f:\ProgramDirectory\iecho    f:\DataDirectory\NeutralFile.cii
```

where:

`f:\ProgramDirectory` and `f:\DataDirectory` must be changed as appropriate.

The .CII file, is divided into sections which organize the piping data in logical groupings. Section divisions are denoted in the neutral file by the '#$' character sequence found in columns 1 and 2. The token following the **#$** character sequence is a section identifier that is used by CAESAR II for data sequencing purposes, and to aid you in reading the neutral file. For each item listed on the following pages, the necessary FORTRAN format for the input/output is provided. The variables listed below are used in dimensioning arrays.

- N1—Base memory allocation quantity used to set array sizes. For example, if N1=2,000, your neutral file can handle up to 2,000 elements.
- N2—1/2 N1
- N3—1/3 N1
- N4—1/4 N1
- N5—1/5 N1
- N6—N1/13.33

NOTE For information on editing ASCII text, see *ASCII Text File Rules* (page 50).

Basic Element Data

#$ ELEMENTS - Contains integer and real data for each element in the input file. The data is organized as shown below.

1. *Real* values for element "i"
2. *Integer* values for element "i"
3. *Real* values for element "i+1"
4. *Integer* values for element "i+1"

These *real* and *integer* values are stored in arrays that are described below.

A 98-member array (REL) contains the real basic-element data. The REL array is dimensioned (N1,98).

Use the FORTRAN format (2X, 6G13.6) to write the values of the following 53 items on the appropriate nine lines of the neutral file.

1. FROM node number
2. TO node number
3. Delta X
4. Delta Y
5. Delta Z
6. Diameter (value stored here is actual OD)

7. Wall Thickness (actual)
8. Insulation Thickness
9. Corrosion Allowance
10. Thermal Expansion Coefficient #1 (or Temperature #1)
11. Thermal Expansion Coefficient #2 (or Temperature #2)
12. Thermal Expansion Coefficient #3 (or Temperature #3)
13. Thermal Expansion Coefficient #4 (or Temperature #4)
14. Thermal Expansion Coefficient #5 (or Temperature #5)
15. Thermal Expansion Coefficient #6 (or Temperature #6)
16. Thermal Expansion Coefficient #7 (or Temperature #7)
17. Thermal Expansion Coefficient #8 (or Temperature #8)
18. Thermal Expansion Coefficient #9 (or Temperature #9)
19. Pressure #1
20. Pressure #2
21. Pressure #3
22. Pressure #4
23. Pressure #5
24. Pressure #6
25. Pressure #7
26. Pressure #8
27. Pressure #9
28. Elastic Modulus (cold)
29. Poisson’s Ratio
30. Pipe Density
31. Insulation Density
32. Fluid Density
33. Minus Mill Tolerance
34. Plus Mill Tolerance
35. Seam Weld (1=Yes, 0=No)
36. Hydro Pressure
37. Elastic Modulus (Hot #1)
38. Elastic Modulus (Hot #2)
39. Elastic Modulus (Hot #3)
40. Elastic Modulus (Hot #4)
41. Elastic Modulus (Hot #5)

42. Elastic Modulus (Hot #6)
43. Elastic Modulus (Hot #7)
44. Elastic Modulus (Hot #8)
45. Elastic Modulus (Hot #9)
46. "wL" Factor
47. Element Orientation Angle (To End), FEATools Data Only
48. Element Orientation Angle (From End), FEATools Data Only
49. Cladding Thickness
50. Cladding Density
51. Insulation + Cladding Weight/length
52. Refractory Thickness
53. Refractory Density

Non-specified *real* values are assigned a value of 0.0 by this interface.

If the delta coordinates are not specified, they default to zero. If the **To/From** fields are not specified, it is considered an error.

Use the FORTRAN format (7X, I5, 1X, A500) to write the values of the following items on the next lines.

NOTE The value before the string indicates the length of the string.

1. Element Name
2. Line Number
3. Line Color and Line Visibility, using the format (2X, 6G13.6)

An 18-member array (IEL) contains the pointers to the auxiliary data arrays. The IEL array is dimensioned (N1,18).

NOTE At this time, only 15 of the members of this array are used.

Use the FORTRAN format (2X, 6I13) to write the values of the following 15 items on the next three lines of the neutral file.

1. Pointer to **Bend Auxiliary** field. This indicates where in the bend auxiliary array the bend data for the current element can be found.
2. Pointer to **Rigid Element Auxiliary** field.
3. Pointer to **Expansion Joint Auxiliary** field.
4. Pointer to **Restraint Auxiliary** field.
5. Pointer to **Displacement Auxiliary** field.
6. Pointer to **Force/Moment Auxiliary** field.
7. Pointer to **Uniform Load Auxiliary** field.
8. Pointer to **Wind Load Auxiliary** field.
9. Pointer to **Element Offset Auxiliary** field.

10. Pointer to **Allowable Stress Auxiliary** field.
11. Pointer to **Intersection Auxiliary** field.
12. Pointer to **Node Name Auxiliary** field.
13. Pointer to **Reducer Auxiliary** field.
14. Pointer to **Flange Auxiliary** field.
15. Pointer to **Nozzle/Equipment Check Auxiliary** field.

NOTE When there is no auxiliary data of a particular type associated with the current element use a pointer value of zero.

Auxiliary Element Data

#$ AUX_DATA

Contains the auxiliary data corresponding to the elements. This data is arranged in the same order as the IAUXAU array described previously. For example, if IAUXAU(1) contains a 3, then there are three bends in the model, and their data is found next in the neutral file. Likewise, if IAUXAU(2) contains a 5, then there are five rigid elements in the model and their data follows the bend data.

Each set of auxiliary data is separated by a sub-section header. If a particular value in IAUXAU is zero, then only the subsection header is written to the neutral file.

The data storage for these arrays is allocated at runtime based on the available free system memory. These arrays are allocated proportionally as a percentage of the n-number of elements allowed. Four proportions are used: 1/2, 1/3, 1/4, and 1/5. These proportions correspond to the variables: N2, N3, N4, and N5. Maintain these proportions to ensure that the neutral file reader can accept the file.

#$ NODENAME

Defines the Node Name data. To maintain downward compatibility, this section is optional. The data for each element set of node names in the input file is listed here.

A two-member array (NAM) defines each set of node names. The NAM array is dimensioned (N6, 2). Use the FORTRAN format (2X, A10, 16X, A10) to read the character name of the **FROM** node and then that of the **TO** node.

#$ BEND

Defines the bend data. The data for each bend in the input file is listed here.

A 15-member array (BND) defines each bend. The BND array is dimensioned (N3,15).

Only 13 items are currently used. Use the FORTRAN format (2X, 6G13.6) to write the values of the following 13 items on the next three lines of the neutral file.

1. Bend radius
2. Type: 1 - single flange; 2 - double flange; 0 or blank - welded
3. Angle to node position #1
4. Node number at position #1
5. Angle to node position #2
6. Node number at position #2

7. Angle to node position #3
8. Node number at position #3
9. Number of miter cuts
10. Fitting thickness of bend if different from the pipe
11. Seam Weld (1=Yes, 0=No)
12. Bend flexibility (K) factor
13. Weld strength reduction factor WL
14. Overlay Thickness

#$ RIGID

Defines the rigid data. The data for each rigid in the input file is listed here.

A single-element array (RIG) is used for each rigid. The RIG array is dimensioned (N3,1). The single element of the array represents the rigid weight.

Use the FORTRAN format (2X, 6G13.6) to write the value.

#$ EXPJT

Defines the expansion joint data. The data for each expansion joint in the input file is listed here. The EXP array is dimensioned (N5,5).

Use FORTRAN format (2X, 6G13.6) to write the values of the following five items on the next line of the neutral file.

1. Axial stiffness
2. Transverse stiffness
3. Bending stiffness
4. Torsional stiffness
5. Effective inside bellows diameter

#$ RESTRANT

Defines the restraint data.

The data for each restraint auxiliary data block in the input file is listed here. The RES array is dimensioned (N2,36).

Use the FORTRAN format (2X, 6G13.6) to write the values of the following nine items on the next two lines of the neutral file. These nine items are repeated six times for the six possible restraints defined in the auxiliary data block.

This requires two lines in the neutral file for each restraint specification. This means twelve lines total for each restraint auxiliary.

1. Restraint node number
2. Restraint type (see additional notes to follow)
3. Restraint stiffness
4. Restraint gap
5. Restraint friction coefficient
6. Restraint connecting node

7. X direction cosine
8. Y direction cosine
9. Z direction cosine

The restraint type is an integer value whose valid range is from 1 to 62. The 62 possible restraint types include:

1 - 5	ANC	X	Y	Z	RX
6 - 10	RY	RZ	GUI	LIM	XSNB
11 - 15	YSNB	ZSNB	+X	+Y	+Z
16 - 20	-X	-Y	-Z	+RX	+RY
21 - 25	+RZ	-RX	-RY	-RZ	+LIM
26 - 30	-LIM	XROD	YROD	ZROD	+XROD
31 - 35	+YROD	+ZROD	-XROD	-YROD	-ZROD
36 - 40	X2	Y2	Z2	RX2	RY2
41 - 45	RZ2	+X2	+Y2	+Z2	-X2
46 - 50	-Y2	-Z2	+RX2	+RY2	+RZ2
51 - 55	-RX2	-RY2	-RZ2	XSPR	YSPR
56 - 60	ZSPR	+XSNB	+YSNB	+ZSNB	-XSNB
61 - 62	-YSNB	-ZSNB			

Use the FORTRAN format (7X, I5, 1X, A100) to write the values of the following two items on the next two lines of the neutral file. These two items follow the restraint specifications above and the neutral file repeats the lines six times for the six possible restraint degrees of freedom defined in the Restraints Auxiliary data in the piping input.

1. Support Tag
2. Support GUID

NOTE The value at the beginning of each string indicates the length of the string. If the string is blank, the value is 0.

#$ DISPLMNT

Defines the displacement data. The data for each displacement auxiliary data block in the input file is listed here.

Use the FORTRAN format (2X, 6G13.6) to write the values of the following 55 items on the next lines of the neutral file. The DIS array is dimensioned (N3,110).

This requires 10 lines in the neutral file for each displacement specification. This means 20

lines total for each displacement auxiliary.

	1) first displacement node number					
{vector 1}	2) DX	3) DY	4) DZ	5) RX	6) RY	7) RZ
{vector 2}	8) DX	9) DY	10) DZ	11) RX	12) RY	13) RZ
{vector 3}	14) DX	15) DY	16) DZ	17) RX	18) RY	19) RZ
{vector 4}	20) DX	21) DY	22) DZ	23) RX	24) RY	25) RZ
{vector 5}	26) DX	27) DY	28) DZ	29) RX	30) RY	31) RZ
{vector 6}	32) DX	33) DY	34) DZ	35) RX	36) RY	37) RZ
{vector 7}	38) DX	39) DY	40) DZ	41) RX	42) RY	43) RZ
{vector 8}	44) DX	45) DY	46) DZ	47) RX	48) RY	49) RZ
{vector 9}	50) DX	51) DY	52) DZ	53) RX	54) RY	55) RZ

	56) second displacement node number					
{vector 1}	57) DX	58) DY	59) DZ	60) RX	61) RY	62) RZ
{vector 2}	63) DX	64) DY	65) DZ	66) RX	67) RY	68) RZ
{vector 3}	69) DX	70) DY	71) DZ	72) RX	73) RY	74) RZ
{vector 4}	75) DX	76) DY	77) DZ	78) RX	79) RY	80) RZ
{vector 5}	81) DX	82) DY	83) DZ	84) RX	85) RY	86) RZ
{vector 6}	87) DX	88) DY	89) DZ	90) RX	91) RY	92) RZ
{vector 7}	93) DX	94) DY	95) DZ	96) RX	97) RY	98) RZ
{vector 8}	99) DX	100) DY	101) DZ	102) RX	103) RY	104) RZ
{vector 9}	105) DX	106) DY	107) DZ	108) RX	109) RY	110) RZ

These 55 items are repeated twice for the two possible displacements defined on the auxiliary.

NOTE Unspecified displacement values (free-displacement degrees of freedom, for example) are designated by using a value of 9999.99.

#$ FORCMNT

Defines the start of the force/moment data. The data for each force/moment auxiliary data block in the input file is listed here.

Use the FORTRAN format (2X, 6G13.6) to write the values of the following 55 items on the next ten lines of the neutral file. The FOR array is dimensioned (N3,38).

This requires ten lines in the neutral file for each force/moment specification. This means 20 lines total for each force/moment auxiliary data block.

	1) first force/moment node number					
{vector 1}	2) FX	3) FY	4) FZ	5) MX	6) MY	7) MZ
{vector 2}	8) FX	9) FY	10) FZ	11) MX	12) MY	13) MZ
{vector 3}	14) FX	15) FY	16) FZ	17) MX	18) MY	19) MZ
{vector 4}	20) FX	21) FY	22) FZ	23) MX	24) MY	25) MZ
{vector 5}	26) FX	27) FY	28) FZ	29) MX	30) MY	31) MZ
{vector 6}	32) FX	33) FY	34) FZ	35) MX	36) MY	37) MZ
{vector 7}	38) FX	39) FY	40) FZ	41) MX	42) MY	43) MZ
{vector 8}	44) FX	45) FY	46) FZ	47) MX	48) MY	49) MZ
{vector 9}	50) FX	51) FY	52) FZ	53) MX	54) MY	55) MZ

	56) second displacement node number					
{vector 1}	57) FX	58) FY	59) FZ	60) MX	61) MY	62) MZ
{vector 2}	63) FX	64) FY	65) FZ	66) MX	67) MY	68) MZ
{vector 3}	69) FX	70) FY	71) FZ	72) MX	73) MY	74) MZ
{vector 4}	75) FX	76) FY	77) FZ	78) MX	79) MY	80) MZ
{vector 5}	81) FX	82) FY	83) FZ	84) MX	85) MY	86) MZ
{vector 6}	87) FX	88) FY	89) FZ	90) MX	91) MY	92) MZ
{vector 7}	93) FX	94) FY	95) FZ	96) MX	97) MY	98) MZ
{vector 8}	99) FX	100) FY	101) FZ	102) MX	103) MY	104) MZ
{vector 9}	105) FX	106) FY	107) FZ	108) MX	109) MY	110) MZ

#$ UNIFORM

Defines the start of the uniform load data. The data for each uniform load in the input file is listed here.

Use FORTRAN format (2X, 6G13.6) to write the values of the following 36 items on the next six lines of the neutral file. The UNI array is dimensioned (N5,36). Currently three vectors of four values each (three directions and a G-load flag) are used.

This requires six lines in the neutral file for each uniform load auxiliary data block.

G-flag is 1 for the input values in G's and 0 for input values in force-per-length notation.

{vector 1 & 2}	UX1	UY1	UZ1	G-flag1	UX2	UY2
{vector 2 & 3}	UZ2	G-flag2	UX3	UY3	UZ3	G-flag3

#$ WIND

Defines the start of the wind/wave data. The data for each wind/wave specification in the input file is listed here. The WIND array is dimensioned (N5,6).

Use the FORTRAN format (2X, 6G13.6) to write the set of values on the next line of the neutral file. This requires a single line in the neutral file for each wind auxiliary.

The data items on each line are as follows:

1. Entry type (0.0 for Wind, 1.0 for Wave, 2.0 for Off)
2. Wind shape factor or wave drag coefficient
3. Wave added mass coefficient
4. Wave lift coefficient
5. Wave marine growth

6. Marine growth density

#$ OFFSETS

Defines the start of the element offset data. The data for each offset pipe in the input file is listed here.

Use FORTRAN format (2X, 6G13.6) to write the values of the following six items on the next line of the neutral file. The OFF array is dimensioned (N5,6).

This requires a single line in the neutral file for each offset auxiliary.

1. Element FROM node offset in X direction
2. Element FROM node offset in Y direction
3. Element FROM node offset in Z direction
4. Element TO node offset in X direction
5. Element TO node offset in Y direction
6. Element TO node offset in Z direction

#$ ALLOWBLS

Defines the start of the allowable stress data. The data for each allowable spec in the input file is listed here.

Use FORTRAN format (2X, 6G13.6) to write the values of the following 153 items on the next 26 lines of the neutral file. The ALL array is dimensioned (N5,153).

1. Cold allowable stress
2. Hot allowable for thermal case #1
3. Hot allowable for thermal case #2
4. Hot allowable for thermal case #3
5. Code cyclic reduction factor for thermal case #1
6. Code cyclic reduction factor for thermal case #2
7. Code cyclic reduction factor for thermal case #3
8. Eff.
9. Sy
10. Fac
11. Pmax
12. Piping code id
13. Hot allowable for thermal case #4
14. Hot allowable for thermal case #5
15. Hot allowable for thermal case #6
16. Hot allowable for thermal case #7
17. Hot allowable for thermal case #8
18. Hot allowable for thermal case #9

19. Code cyclic reduction factor for thermal case #4
20. Code cyclic reduction factor for thermal case #5
21. Code cyclic reduction factor for thermal case #6
22. Code cyclic reduction factor for thermal case #7
23. Code cyclic reduction factor for thermal case #8
24. Code cyclic reduction factor for thermal case #9

Items 25 through 32 represent Cycles, and items 33 through 40 represent Stresses for eight BW (butt-weld)/Class D Fatigue pairs.

Items 41 through 48 represent Cycles, and items 49 through 56 represent Stresses for eight FW (fillet-weld)/Class E Fatigue pairs.

Items 57 through 64 represent Cycles, and items 65 through 72 represent Stresses for eight Class F Fatigue pairs used with TD/12 piping code.

Items 73 through 80 represent Cycles, and items 81 through 88 represent Stresses for eight Class G Fatigue pairs used with TD/12 piping code.

Items 89 through 96 represent Cycles, and items 97 through 104 represent Stresses for eight Class W Fatigue pairs used with TD/12 piping code.

Item 105 – Elastic Modulus correction

Item 106 – has different meanings based on the active piping code: Allowed Cycles Maximum (per B31.3); Restrained Piping (per B31.8); Material Composition/Type (per HPGSL and JPI).

Item 107 – UTS ambient

Item 108 – Allowable Sy/St value

Items 109 through 117 represent nine SY values at temperature.

Items 118 through 126 represent nine UTS values at temperature.

ISO 14692-2017 (items 130-136) – A0 (130), A2, Buried check box, Ehb, Shape factor Df, vertical deflection y/D (135), Curve radius R (136).

DNV-2017 (items 137-149)- Alpha:h (137), Alpha:fab, Supplentary Req, Material LP, Burst Ope, Burst Test, Collapse, Prop Buckling, LCC, DCC, Gamma:c, R, Condition.

Items 127 through 153 are currently unused. Write the value of 0.000000.

NOTES

- Some of these items (notably 8-24) may have various meanings based on the active piping code.
- Piping code ISO-14692 has special mapping for the first 24 items.

#$ SIF&TEES

Defines the start of the SIF/TEE data. The data for each SIF/TEE specification in the input file is listed here.

Use FORTRAN format (2X, 6G13.6) to write the values of the following 30 items, for each of the two tees that can be specified on the dialog box. The SIF array is dimensioned (N4,60).

The software requires five lines in the neutral file for each SIF/TEE specified. This means you must specify 10 lines total for each auxiliary element.

NOTE The information in parenthesis below gives information about each input value.

1. Node (the intersection node number)
2. Type (the intersection type code, if not specified this auxiliary this is only used to specify SIFs)
3. In-Plane SIF (i_i)*
4. Out-Plane SIF (i_o)*
5. Weld (d) (circumferential weld mismatch, used for butt welds and tapered transitions
6. Fillet (fillet leg length)
7. Pad Thk (thickness of the reinforcing pad)
8. Ftg Ro (fitting outside radius for branch connections)
9. Crotch R (crotch radius of the formed lip on an extruded welding tee)
10. Weld ID (weld ID value)
11. B1 (code-specific value)
12. B2 (code-specific value)
13. In-Plane Index (I_i)*
14. Out-Plane Index (I_o)*
15. Torsion SIF (i_t)*
16. Axial SIF (i_a)*
17. Pressure SIF (i_p)*
18. Pressure Index (I_p)*
19. 19-22*

23.(code-related "Note" options)

24. (code-related "Note" options)

25. Axial Index (I_a)

26. Torsion Index (I_t)

NOTES

- * Values 3, 4, 11, and 13-22 are for the IGE/TD/12 piping code.
- Some of these values may have different meanings based on the piping code you have selected. For more information on piping input specifics, see *SIFs & Tees* (page 142).

#$ REDUCERS

This subsection header defines the start of the REDUCER data. The data for each REDUCER spec in the input file is listed here.

Use FORTRAN format (2X, 6G13.6) to write the values of the following five items on the next line of the neutral file. The RED array is dimensioned (N6,5).

This requires one line in the neutral file for each REDUCER specified.

1. Second diameter of the reducer

2. Second thickness of the reducer
3. Alpha angle of the reducer
4. R1 value of the reducer for the TD/12 piping code
5. R2 value of the reducer for the TD/12 piping code

These values are repeated for the second intersection specification.

#$ FLANGES

Defines the FLANGE data. The data for each FLANGE spec in the input file is listed here. There are 72 data values used to describe a flange.

1. FROM/TO (0 = FROM, 1 = TO, 2= BOTH)
2. METHOD (0 = PEQ, 1 = ASME NC)
3. GASKET OR BOLT CIRCLE DIAMETER, DEPENDING ON METHOD
4. BOLT AREA (ASME METHOD ONLY)
5. SYC (ASME METHOD ONLY)
6. SY1 (ASME METHOD ONLY)
7. SY2 (ASME METHOD ONLY)
8. SY3 (ASME METHOD ONLY)
9. SY4 (ASME METHOD ONLY)
10. SY5 (ASME METHOD ONLY)
11. SY6 (ASME METHOD ONLY)
12. SY7 (ASME METHOD ONLY)
13. SY8 (ASME METHOD ONLY)
14. SY9 (ASME METHOD ONLY)
15. 15-24 CLASS NAME, (40 CHAR MAX)
16. 25-48 24 TEMPERATURES OF THE TEMP/PRESS RATING CURVE (PEQ METHOD ONLY)
17. 49-72 24 PRESSURES OF THE TEMP/PRESS RATING CURVE (PEQ METHOD ONLY)

These values are arranged in the neutral file on 12 lines using a format of (2X, 6G13.6) unless otherwise specified:

Line 1: Flange items 1-5

Line 2: Flange items 6-11

Line 3: Flange items 12-14

Line 4: Class Name, using a format of (2X, A40)

Line 5: Flange items 25-30

Line 6: Flange items 31-36

Line 7: Flange items 37-42

Line 8: Flange items 43-48

Line 9: Flange items 49-54

Line 10: Flange items 55-60

Line 11: Flange items 61-66

Line 12: Flange items 67-72

All 12 lines must be written to the neutral file for each flange. Unused fields/values can be represented by 0.00.

#$ EQUIPMNT

Defines the Equipment/Nozzle Check data. The data for each EQUIPMNT spec in the input file is listed here. There are two sets of 17 data values that use a format of (2X, 6G13.6) as shown below:

1. Node Number
2. Limiting load value FX
3. Limiting load value FY
4. Limiting load value FZ
5. Limiting load value MX
6. Limiting load value MY
7. Limiting load value MZ
8. Reference axis direction cosine CosX
9. Reference axis direction cosine CosY
10. Reference axis direction cosine CosZ
11. Flange rating
12. Interaction method: 0=absolute; 1=SRSS; 2=Unity Check;

Items 13-17 are spares represented by 0.00. These values are arranged in the neutral file on six lines. All six lines must be written to the neutral file for each **Nozzle/Equipment** check. Unused fields/values can be represented by 0.00.

Miscellaneous Data Group #1

#$ MISCEL_1

Contains the material ID (RRMAT) for each element in the input file, the nozzle data (VFLEX), the hanger data, and the execution options.

Material ID

Contains the material ID number in the first array for each element in the input file. Use the FORTRAN format (2X, 6G13.6).

The RRMAT array is dimensioned (N1). The material IDs range from 1 to 699 (See the User's Guide for details).

The number of lines required to write the RRMAT array in the neutral file is determined by the following FORTRAN routine:

```
NLINES = NUMELT / 6
```

```
IF(MOD(NUMELT,6).NE.0)THEN
NLINES = NLINES + 1
ENDIF
```

Nozzles

Describes the flexible WRC-297, PD 5500, API 650, and Custom nozzles in the input file.

NOTE The value 9999.99 represents infinity or an undefined term.

Use the FORTRAN format (2X, 6G13.6). The nozzle (VFLEX) contains 22 values for each nozzle in the input. This requires four lines in the neutral file.

For WRC-297 nozzles, the items are:

1. Nozzle node number
2. Vessel node number (optional)
3. Nozzle type indicator (0 = WRC-297)
4. Nozzle outside diameter (in.)
5. Nozzle wall thickness (in.)
6. Vessel outside diameter (in.)
7. Vessel wall thickness (in.)
8. Vessel reinforcing pad thickness (in.)
9. Spare (not used)
10. Dist. to stiffeners or head (in.)
11. Dist. to opposite side stiffeners or head (in.)
12. Vessel centerline direction vector X
13. Vessel centerline direction vector Y
14. Vessel centerline direction vector Z
15. Vessel temperature (optional) (°F)
16. Vessel material # (optional)
17. Spare (not used)
18. Spare (not used)
19. Spare (not used)
20. Spare (not used)
21. Spare (not used)
22. Spare (not used)

For API 650 nozzles, the items are:

1. Nozzle node number
2. Tank node number (optional)
3. Nozzle type indicator (1.0 = API-650)
4. Nozzle outside diameter (in.)

5. Nozzle wall thickness (in.)
6. Tank outer diameter (in.)
7. Tank wall thickness (in.)
8. Spare (not used)
9. Reinforcing on shell (1) or nozzle (2)
10. Nozzle height (in.)
11. Fluid height (in.)
12. Displacement vector (integer value between 1 and 9, 0 if unused)
13. Specific gravity of fluid
14. Tank thermal expansion coefficient (in/in/deg)
15. Delta temperature (°F)
16. Elastic modulus (psi)
17. Spare (not used)
18. Spare (not used)
19. Spare (not used)
20. Spare (not used)
21. Spare (not used)
22. Spare (not used)

For PD 5500 nozzles, the items are:

1. Nozzle node number
2. Vessel node number (optional)
3. Nozzle type indicator (.0 = PD 5500)
4. Nozzle outside diameter (in.)
5. Vessel type (0-cylinder, 1-sphere)
6. Vessel outside diameter (in.)
7. Vessel wall thickness (in.)
8. Vessel reinforcing pad thickness (in.)
9. Spare (not used)
10. Dist. to stiffeners or head (in.)
11. Dist. to opposite side stiffeners or head (in.)
12. Vessel centerline direction vector X
13. Vessel centerline direction vector Y
14. Vessel centerline direction vector Z
15. Vessel Temperature (optional) (°F)
16. Vessel Material # (optional)

17. Spare (not used)
18. Spare (not used)
19. Spare (not used)
20. Spare (not used)
21. Spare (not used)
22. Spare (not used)

For Custom nozzles, the items are:

1. Nozzle node number
2. Vessel node number (optional)
3. Nozzle type indicator (3.0 = Custom)
4. Nozzle outside diameter (in.)
5. Nozzle wall thickness (in.)
6. Vessel outside diameter (in.)
7. Vessel wall thickness (in.)
8. Vessel reinforcing pad thickness (in.)
9. Spare (not used)
10. Spare (not used)
11. Spare (not used)
12. Vessel centerline direction vector X
13. Vessel centerline direction vector Y
14. Vessel centerline direction vector Z
15. Spare (not used)
16. Vessel material # (optional)
17. Spare (not used)
18. Spare (not used)
19. Axial stiffness (lb.\in.)
20. In-plane stiffness (in.-lb./deg.)
21. Out-of-plane stiffness (in.-lb./deg.)
22. Torsional stiffness (in.-lb./deg.)

Hangers

Describes the spring hangers in the input file. Some of the hanger data listed below represents uninitialized data. In the instances where this uninitialized data represent infinite values (such as maximum travel limit and available space), it is reported here as 9999.99.

The next line contains values for the following parameters in FORTRAN format (2X, I13, 5G13.6):

- IDFTABLE is the default hanger table.

- DEFVAR is the default for allowed load variation.
- DEFRIG is the default for rigid support displacement criteria.
- DEFMXTRAVEL is the default for maximum allowed travel.
- DEFSHTSPR is the default for allowing short range springs (0=no 1=yes).
- DEFMUL is the default multi-load case design option.

The next line contains values for the following parameters in the FORTRAN format (2X, 5I13):

- IDFOPER is the default number of hanger design operating cases (always 1).
- IACTCLD is the default cold load calculation switch (0=no, 1=yes).
- IHGRLDS is the number of hanger operating loads (0 -3).
- IACTUAL is the load case defining actual cold loads.
- IMULTIOPTS is the multi-load case design option (1-7).

An array of hanger node numbers (IHGRNODE) is read and written for each hanger in the input file and is dimensioned (N5). There are seven lines in the neutral file for this data if all N5 hangers are specified. Use the FORTRAN format (2X, 6I13).

The neutral file reads and writes an 11-element array (HGRDAT) for each hanger in the input file. The HGRDAT array is dimensioned (11,N5). Each hanger in the model requires two lines in the neutral file. Use the FORTRAN format (2X, 6G13.6).

1. Hanger stiffness
2. Allowable load variation
3. Rigid support displacement criteria
4. Allowed space for hanger
5. Cold load #1 (theoretical)
6. Hot load #1 (initialize to 0.0)
7. User defined operating load f/ variable springs (init to 0.0)
8. Maximum allowed travel limit
9. Multiple load case design option
10. Hanger hardware weight
11. Constant effort support load (CEFF)

Use the FORTRAN format (7X, I5, 1X, A100) to write the values of the following two items on the next two lines of the neutral file. These two items follow the hanger specifications above and the neutral file repeats the lines for each hanger defined in the Hangers Auxiliary data in the piping input.

1. Hanger Tag
2. Hanger GUID

NOTE The value at the beginning of each string indicates the length of the string. If the string is blank, the value is 0.

A four-element array (IHGRFREE) is read/written for each hanger in the input file. The IHGRFREE array is dimensioned (4,N5). Each hanger in the file requires one line in the neutral file.

Use FORTRAN format (2X, 6I13).

1. Anchor node to be freed (#1)
2. Anchor node to be freed (#2)
3. d.o.f. type for #1 (1-free Y, 2-free XY, 3-free ZY, 4-free X, Y, Z, 5-free all)
4. d.o.f. type for #2

An array (IHGRNUM) lists the number of hangers at this location for each hanger in the input file. There is one entry here for every hanger in the file. The IHGRNUM array is dimensioned (N5). There are seven lines in the neutral file for this data if all N5 hangers are specified. Use the FORTRAN format (2X, 6I13).

An array (IHGRTABLE) lists the hanger table numbers for each hanger in the input file. There is one entry here for every hanger in the file. The IHGRTABLE is dimensioned (N5). There are seven lines in the neutral file for this data if all N5 hangers are specified. Use the FORTRAN format (2X, 6I13).

An array of flags (IHGRSHORT) indicates if short range springs can be used at each hanger location. The IHGRSHORT array is dimensioned (N5). There are seven lines in the neutral file for this data. Use the FORTRAN format (2X, 6I13).

- 0 = cannot use short range springs
- 1 = can use short range springs

An array of connecting node numbers (IHGRCN) is available for each hanger. The IHGRCN array is dimensioned (N5). There are seven lines in the neutral file for this data if all N5 hangers are specified. Use the FORTRAN format (2X, 6I13).

Execution Options

Defines the execution options used by CAESAR II. Use the FORTRAN format (2X, 4I13, G13.6, I13). This requires three lines in the neutral file. These values are:

- Print forces on rigids and expansion joints 0=no, 1=yes
- Print alphas & pipe props. during error checking 0=no, 1=yes
- Activate Bourdon Pressure Effects 0, 1, or 2
- Activate Branch Error and Coordinate Prompts 0=no, 1=yes
- Thermal Bowing Delta Temperature degrees
- Use Liberal Stress Allowable 0=no, 1=yes

For the following data, use the FORTRAN format: (2X, I13, 2G13.6, 3I13):

- Uniform Load Input in g’s 0=no, 1=yes
- Stress Stiffening due to Pressure 0, 1, 2
- Ambient Temperature (If not 70.00 deg F) degrees
- FRP Expansion * 1,000,000 len/len/deg
- Optimizer 0-Both, 1-CuthillMcKee, 2-Collins

- Next Node Selection 0-Decreasing, 1-Increasing

For the following data, use the FORTRAN format (2X, 4I13, G13.6, I13):

- Final Ordering 0-Reversed, 1-Not Reversed
- Collins Ordering 0-Band, 1-No. of Coefficients
- Degree Determination 0-Connections, 1-Band
- User Control 0-None, 1-Allow User Re-Looping
- FRP Shear ratio
- Laminate type

For the following data, use the FORTRAN format (2X, I13):

- North arrow

Units Conversion Data

#$ UNITS

Defines both the conversion constants and the conversion labels. The conversion constants are all **REAL*4** values in FORTRAN format (2X, 6G13.6). This requires four lines in the neutral file. The character definitions for the labels are listed below.

CNVLEN

Defines the length conversion

CNVFOR

Defines the force conversion

CNVMAS

Defines the mass conversion

CNVMIN

Defines the moment (input) conversion

CNVMOU

Defines the moment (output) conversion

CNVSTR

Defines the stress conversion

CNVTSC

Defines the temperature conversion

CNVTOF

Defines the temperature offset

CNVPRE

Defines the Pressure conversion

CNVYM

Defines the Young’s modulus conversion

CNVPDN

Defines the pipe density conversion

CNVIDN

Defines the insulation density conversion

CNVFDN

Defines the fluid density conversion

CNVTSF

Defines the translational stiffness conversion

CNVUNI

Defines the uniform load conversion

CNVWND

Defines the wind load conversion

CNVELE

Defines the elevation conversion

CNVCLN

Defines the compound length conversion

CNVDIA

Defines the diameter conversion

CNVTHK

Wall thickness conversion

Next, enter the following labels for units, one per line, in the format given in the label descriptions. This requires 24 lines in the neutral file.

CCVNAME

Defines the name of the units used, such as English (CHARACTER*15)

CCVNOM

Sets On or Off, and tells PREPIP whether or not nominal diameters are allowed (CHARACTER* 3)

CCVLEN

Defines the length label (CHARACTER* 3)

CCVFOR

Defines the force label (CHARACTER* 3)

CCVMAS

Defines the mass label (CHARACTER* 3)

CCVMIN

Defines the moment (input) label (CHARACTER* 6)

CCVMOU

Defines the moment (output) label (CHARACTER* 6)

CCVSTR

Defines the stress label (CHARACTER*10)

CCVTSC

Defines the temperature label (CHARACTER* 1)

CCVTOF

Defines the temperature offset/label (CHARACTER* 1)

CCVPRE

Defines the pressure label (CHARACTER*10)

CCVYM

Defines Young's modulus label (CHARACTER*10)

CCVPDN

Defines the pipe density label (CHARACTER*10)

CCVIDN

Defines the insulation density label (CHARACTER*10)

CCVFDN

Defines the fluid density label (CHARACTER*10)

CCVTSF

Defines the translational stiffness label (CHARACTER* 7)

CCVRSF

Defines the rotational stiffness label (CHARACTER*10)

CCVUNI

Defines the uniform load label (CHARACTER* 7)

CCVGLD

Defines the gravitational load label (CHARACTER* 3)

CCVWND

Defines the wind load label (CHARACTER*10)

CCVELE

Defines the elevation label (CHARACTER* 3)

CCVCLN

Defines the compound length label (CHARACTER* 3)

CCVDIA

Defines the diameter label (CHARACTER* 3)

CCVTHK

Defines the wall thickness label (CHARACTER* 3)

Nodal Coordinate Data

#$ COORDS

Specifies the X, Y, Z global coordinates of the starting node point of each discontinuous piping segment. The data in this section of the neutral file is optional; it may not exist. The existence of this data depends on user preference and the particular job. The data is defined below.

NXYZ

Defines how many sets of coordinates follow. Use FORTRAN format (2X, I13).

INODE, XCORD, YCORD, ZCORD

Consists of four values in a line and is repeated NXYZ times.

Use FORTRAN format (2X, I13, 3F13.4) to define a node number and the X, Y, Z global coordinates.

NOTE This section only exists in Versions 3.22 and later.

Version and Job Title Information

#$ VERSION - Provides section header information.

Use FORTRAN format (2X, 2G13.6, I8) to write the values of the following variables on the first line of the neutral file:

- GVERSION is the version of the neutral file interface being used. This corresponds to the major version number of CAESAR II (4 for 4.x., for example).
- RVERSION is the specific CAESAR II version generating this file, for example 4.50.
- ICODEPAGE is the Windows (ANSI) code page identifier, for example 1252.

The next 60 lines of 75 characters each are reserved for the CAESAR II title-page text. Use FORTRAN format (2X, A75).

The last line of the job title array, if blank, is set by this transfer interface. The text that is set here indicates that the file was created by the CAESAR II interface.

Control Information

#$ CONTROL

Provides the section division header. The **#$** and space are required, and the word **CONTROL** is in all uppercase.

Use the FORTRAN format (2X, 6I13) to write the values of the following variables on the next line of the neutral file:

NUMELT

Defines the number of piping elements (spreadsheets) in the input file.

NUMNOZ

Defines the number of nozzles in the input file.

NOHGRS

Defines the number of spring hangers in the input file.

NONAM

Defines the number of Node Name data blocks in the input file.

NORED

Defines the number of reducers in the input file.

NUMFLG

Defines the number of flanges in the input file.

Write 13 items that contain the number of auxiliary data types used in the input file followed by the vertical axis indicator. Use the FORTRAN format (2X, 6I13). These 13 values are:

- The number of bend auxiliary data blocks in the input file.
- The number of rigid-element auxiliary data blocks in the input file.
- The number of expansion-joint auxiliary data blocks in the input file.
- The number of restraint auxiliary data blocks in the input file.
- The number of displacement auxiliary data blocks in the input file.
- The number of force/moment auxiliary data blocks in the input file.
- The number of uniform-load auxiliary data blocks in the input file.
- The number of wind-load auxiliary data blocks in the input file.
- The number of element-offset auxiliary data blocks in the input file.
- The number of allowable-stress auxiliary data blocks in the input file.
- The number of intersection auxiliary data blocks in the input file. IZUP flag. Equal to zero (0) for the global -Y axis vertical; equal to 1 for the global -Z axis vertical.
- The number of (nozzle) equipment limits data blocks in the input file.

Neutral File Generator

Generates the CAESAR II neutral file (.cii), which you use to send data to or from the standard CAESAR II input file to another interface or a third-party application.

Conversion Type

Select the type of file you want to convert using the Neutral File Generator. You can convert a neutral file (.cii) to an CAESAR II input file (._a) or vice-versa.

Next, browse and select or type the name of the neutral file to be converted and click **Convert**. The software prompts you when the conversion completes. Click **OK** and indicate whether you want to perform another file conversion. When you are done converting files, click **No** and the software exits from the dialog box.

CAESAR II Input Export Options

Specifies the input options to export. The options you select on this page determine what columns appear in the specified exported format.

CAESAR II exports the input and results to the format selected on **CAESAR II Input and Output Files** (page 1204).

Finish

Exports input data for the selected CAESAR II file. This option only displays when you clear **Export Output Data Also** (page 1204).

Elements (Input Export Options)

Exports the basic element data including pointers to auxiliary data.

Microsoft Access table name: INPUT_BASIC_ELEMENT_DATA
XML Primary Tag: PIPINGELEMENT.

Data Name	Access Column Name	XML Tag Name
Element identifier	ELEMENTID	
Element name	ELEMENT_NAME	ELEMENT_NAME
From node number	FROM_NODE	FROM_NODE
To node number	TO_NODE	TO_NODE
From node name	FROM_NODE_NAME	
To node name	TO_NODE_NAME	
Line number	LINE_NO	
Delta X	DELTA_X	DELTA_X
Delta Y	DELTA_Y	DELTA_Y
Delta Z	DELTA_Z	DELTA_Z
Diameter (actual OD)	DIAMETER	DIAMETER
Wall thickness (actual)	WALL_THICK	WALL_THICK
Insulation thickness	INSUL_THICK	INSUL_THICK
Corrosion allowance	CORR_ALLOW	CORR_ALLOW

Data Name	Access Column Name	XML Tag Name
Thermal Expansion Coefficient #1 or Temperature #1	TEMP_EXP_C1	TEMP_EXP_C1
Thermal Expansion Coefficient #2 or Temperature #2	TEMP_EXP_C2	TEMP_EXP_C2
Thermal Expansion Coefficient #3 or Temperature #3	TEMP_EXP_C3	TEMP_EXP_C3
Thermal Expansion Coefficient #4 or Temperature #4	TEMP_EXP_C4	TEMP_EXP_C4
Thermal Expansion Coefficient #5 or Temperature #5	TEMP_EXP_C5	TEMP_EXP_C5
Thermal Expansion Coefficient #6 or Temperature #6	TEMP_EXP_C6	TEMP_EXP_C6
Thermal Expansion Coefficient #7 or Temperature #7	TEMP_EXP_C7	TEMP_EXP_C7
Thermal Expansion Coefficient #8 or Temperature #8	TEMP_EXP_C8	TEMP_EXP_C8
Thermal Expansion Coefficient #9 or Temperature #9	TEMP_EXP_C9	TEMP_EXP_C9
Pressure #1	PRESSURE1	PRESSURE1
Pressure #2	PRESSURE2	PRESSURE2
Pressure #3	PRESSURE3	PRESSURE3
Pressure #4	PRESSURE4	PRESSURE4
Pressure #5	PRESSURE5	PRESSURE5
Pressure #6	PRESSURE6	PRESSURE6
Pressure #7	PRESSURE7	PRESSURE7
Pressure #8	PRESSURE8	PRESSURE8
Pressure #9	PRESSURE9	PRESSURE9
Elastic Modulus	MODULUS	MODULUS
Hot Elastic Modulus #1	HOT_MOD1	

Data Name	Access Column Name	XML Tag Name
Hot Elastic Modulus #2	HOT_MOD2	
Hot Elastic Modulus #3	HOT_MOD3	
Hot Elastic Modulus #4	HOT_MOD4	
Hot Elastic Modulus #5	HOT_MOD5	
Hot Elastic Modulus #6	HOT_MOD6	
Hot Elastic Modulus #7	HOT_MOD7	
Hot Elastic Modulus #8	HOT_MOD8	
Hot Elastic Modulus #9	HOT_MOD9	
Poisson's Ratio	POISSONS	POISSONS
Pipe Density	PIPE_DENSITY	PIPE_DENSITY
Insulation Density	INSUL_DENSITY	INSUL_DENSITY
Fluid Density	FLUID_DENSITY	FLUID_DENSITY
Refractory Thickness	REFRACT_THK	
Refractory Density	REFRACT_DENSITY	
Cladding Thickness	CLAD_THK	
Cladding Density	CLAD_DENSITY	
Insulation and Cladding Unit Weight	INSUL_CLAD_UNIT_WEIGHT	
Material Number	MATERIAL_NUM	MATERIAL_NUM
Material Name	MATERIAL_NAME	MATERIAL_NAME
Plus Mill Tolerance	MILL_TOL_PLUS	MILL_TOL_PLUS
Minus Mill Tolerance	MILL_TOL_MINUS	MILL_TOL_MINUS
Seam Weld	SEAM_WELD	SEAM_WELD
See **Auxiliary Data Pointers** for a list of database pointers to auxiliary data, which also appear in this input options output table.		

Auxiliary Data Pointers

The auxiliary data pointers indicate the location of the details for each piece of auxiliary data. For example, if Bend Pointer is equal to 1 here, then the details of this bend will be contained in the Bend table where the bend number is also equal to 1.

NOTE Auxiliary pointers are only applicable to data export for Microsoft Access and Microsoft Excel and not to XML.

Pointer Type	Access Column Name	XML Tag Name
Bend Array Pointer	BEND_PTR	BEND
Rigid Array Pointer	RIGID_PTR	RIGID
Expansion Joint Array Pointer	EXPJ_PTR	EXPANSIONJOINT
Restraint Array Pointer	REST_PTR	RESTRAINT
Displacement Array Pointer	DISP_PTR	DISPLACEMENTS
Force/Moment Array Pointer	FORCMNT_PTR	FORCEMOMENTS
Uniform Load Array Pointer	ULOAD_PTR	UNIFORMLOAD
Wind/Wave Load Array Pointer	WLOAD_PTR	WIND OR WAVE
Element Offset Array Pointer	EOFF_PTR	OFFSET
Allowable Stress Array Pointer	ALLOW_PTR	ALLOWABLESTRESS
Intersection Array Pointer	INT_PTR	SIF
Hangers Array Pointer	HGR_PTR	HANGER
Nozzles Array Pointer	NOZ_PTR	NOZZLE
Reducers Array Pointer	REDUCER_PTR	REDUCER
Flanges Array Pointer	FLANGE_PTR	FLANGE

Bends (Input Export Options)

Exports all the bend information defined in the job. Below are the details of the bend data available from CAESAR II along with the respective column names and XML tag names.

Microsoft Access table name: INPUT_BENDS
Microsoft Excel section name: BEND DATA
XML Primary Tag: BEND

Data Name	Access Column Name	XML Tag Name
Bend Pointer	BEND_PTR	
Bend Radius	RADIUS	RADIUS
Type (1-Single flange, 2- double flange, 0 or blank- welded)	TYPE	TYPE
Angle to node position #1	ANGLE1	ANGLE1
Node number at position #1	NODE1	NODE1
Angle to node position #2	ANGLE2	ANGLE2
Node number at position #2	NODE2	NODE2
Angle to node position #3	ANGLE3	ANGLE3
Node number at position #3	NODE3	NODE3
Number of miter cuts	NUM_MITER	NUM_MITER
Fitting thickness of bend if different from the pipe	FIT_THICK	FITTINGTHICKNESS
Bend K Factor	KFACTOR	
Is Seam Welded?	SEAM_WELD	
WL Factor	WI_FACTOR	

Access and Excel contain a number that identifies on which element they were defined.

INPUT_BASIC_ELEMENT_DATA.BEND_PTR = INPUT_BENDS.BEND_PTR

Rigids (Input Export Options)

Exports rigid information of all rigid elements defined in the input file.

Microsoft Access table name: INPUT_RIGIDS
Microsoft Excel Section Name: RIGID DATA
XML Primary Tag: RIGID

Data Name	Access Column Name	XML Tag Name
Rigid Pointer	RIGID_PTR	
Rigid Weight	RIGID_WGT	WEIGHT

Access and Excel contain a number that identifies on which element they were defined.

INPUT_BASIC_ELEMENT_DATA.RIGID_PTR = INPUT_BENDS.RIGID_NUM.

Expansion Joints (Input Export Options)

Export expansion joint information of all expansion joints defined in the input file.

Microsoft Access table name: INPUT_EXPJT
Microsoft Excel Section Name: EXPANSION JOINT DATA
XML Primary Tag Name: EXPANSIONJOINT

Data Name	Access Column Name	XML Tag Name
Expansion Joint Pointer	EXPJT_PTR	
Axial stiffness	AXIAL_STIF	AXIAL_STIF
Transverse stiffness	TRANS_STIF	TRANS_STIF
Bending Stiffness	BEND_STIF	BEND_STIF
Torsional stiffness	TORS_STIF	TORS_STIF
Effective inside bellows diameter	BEL_DIA	BEL_DIA

Access and Excel contain a number that identifies on which element they were defined.

INPUT_BASIC_ELEMENT_DATA.EXPJT_PTR = INPUT_EXPJT.EXPJT_PTR.

Restraints (Input Export Options)

Export Restraint information of all restraints defined in the input file.

Microsoft Access Table Name: INPUT_RESTRAINTS
Microsoft Excel Section Name: RESTRAINT DATA
XML Primary Tag Name: RESTRAINT

Data Name	**Column Name**	**XML Tag Name**
Restraint pointer	REST_PTR	
Node number	NODE_NUM	NODE
Node name	NODE_NAME	
Restraint Codes (page 1212)	RES_TYPEID	TYPE
Stiffness	STIFFNESS	STIFFNESS
Gap	GAP	GAP
Friction coefficient	FRIC_COEF	FRIC_COEF
Connecting node	CNODE	CNODE
X direction cosine	XCOSINE	XCOSINE
Y direction cosine	YCOSINE	YCOSINE
Z direction cosine	ZCOSINE	ZCOSINE
Tag	RES_TAG	TAG
GUID	RES_GUID	GUID

Access and Excel contain a number that identifies on which element they were defined.

INPUT_BASIC_ELEMENT_DATA.REST_PTR = INPUT_RESTRAINTS.REST_PTR.

Restraint Codes

There are 62 different restraints available in CAESAR II. When the restraint information is exported to a format such as Microsoft Access, a restraint type code is exported. The following is the key for those restraint codes.

Code	Abbreviation	Type
1	ANC	Anchor
2	X	Translational Double Acting
3	Y	Translational Double Acting
4	Z	Translational Double Acting
5	RX	Rotational Double Acting
6	RY	Rotational Double Acting
7	RZ	Rotational Double Acting
8	GUI	Guide, Double Acting
9	LIM	Double Acting Limit Stop
10	XSNB	Translational Double Acting Snubber
11	YSNB	Translational Double Acting Snubber
12	ZSNB	Translational Double Acting Snubber
13	+X	Translational Directional
14	+Y	Translational Directional
15	+Z	Translational Directional
16	-X	Translational Directional
17	-Y	Translational Directional
18	-Z	Translational Directional
19	+RX	Rotational Directional
20	+RY	Rotational Directional
21	+RZ	Rotational Directional

Code	Abbreviation	Type
22	-RX	Rotational Directional
23	-RY	Rotational Directional
24	-RZ	Rotational Directional
25	+LIM	Directional Limit Stop
26	-LIM	Directional Limit Stop
27	XROD	Large Rotation Rod
28	YROD	Large Rotation Rod
29	ZROD	Large Rotation Rod
30	+XROD	Large Rotation Rod
31	+YROD	Large Rotation Rod
32	+ZROD	Large Rotation Rod
33	-XROD	Large Rotation Rod
34	-YROD	Large Rotation Rod
35	-ZROD	Large Rotation Rod
36	X2	Translational Double Acting Bilinear
37	Y2	Translational Double Acting Bilinear
38	Z2	Translational Double Acting Bilinear
39	RX2	Rotational Double Acting Bilinear
40	RY2	Rotational Double Acting Bilinear
41	RZ2	Rotational Double Acting Bilinear
42	+X2	Translational Directional Bilinear
43	+Y2	Translational Directional Bilinear
44	+Z2	Translational Directional Bilinear
45	-X2	Translational Directional Bilinear

Code	Abbreviation	Type
46	-Y2	Translational Directional Bilinear
47	-Z2	Translational Directional Bilinear
48	+RX2	Rotational Directional Bilinear
49	+RY2	Rotational Directional Bilinear
50	+RZ2	Rotational Directional Bilinear
51	+RX2	Rotational Directional Bilinear
52	+RY2	Rotational Directional Bilinear
53	+RZ2	Rotational Directional Bilinear
54	-RX2	Rotational Directional Bilinear
55	-RY2	Rotational Directional Bilinear
56	-RZ2	Rotational Directional Bilinear
57	+XSNB	Directional Snubber
58	+YSNB	Directional Snubber
59	+ZSNB	Directional Snubber
60	-XSNB	Directional Snubber
61	-YSNB	Directional Snubber
62	-ZSNB	Directional Snubber

Displacements (Input Export Options)

Export user specified displacement information defined in the input file.

Microsoft Access Table Name: INPUT_DISPLMNT
Microsoft Excel Section Name: DISPLACEMENT DATA
XML Primary Tag Name: DISPLACEMENTS

Data Name	Column Name	XML Tag Name
Displacement Pointer	DISP_PTR	
Displacement Number	DISP_NUM	DISP_NUM
Node Number	NODE_NUM	NODE_NUM
Vector Number	VECTOR_NUM	NUMBER*
X axis displacement	DX	DX*
Y axis displacement	DY	DY*
Z axis displacement	DZ	DZ*
X axis rotation	RX	RX*
Y axis rotation	RY	RY*
Z axis rotation	RZ	RZ*

*These tags are child tags of the VECTOR tag.

Access and Excel contain a number that identifies on which element they were defined.

INPUT_BASIC_ELEMENT_DATA.DISP_PTR = INPUT_DISPLMNT.DISP_PTR

Forces/Moments (Input Export Options)

Export user specified forces/moments information defined in the input file.

Microsoft Access Table Name: INPUT_FORCMNT
Microsoft Excel Section Name: FORCES/MOMENTS DATA
XML Primary Tag Name: FORCESMOMENTS

Data Name	Column Name	XML Tag Name
Force/Moment Pointer	FORCMNT_PTR	
Force/Moment Number	FORCMNT_NUM	FORCMNT_NUM
Node Number	NODE_NUM	NODE_NUM

Data Name	Column Name	XML Tag Name
Vector Number	VECTOR_NUM	NUMBER*
X axis force	FX	FX*
Y axis force	FY	FY*
Z axis force	FZ	FZ*
X axis moment	MX	MX*
Y axis moment	MY	MY*
Z axis moment	MZ	MZ*
*These tags are child tags of the VECTOR tag.		

Access and Excel contain a number that identifies on which element they were defined.

INPUT_BASIC_ELEMENT_DATA.FORCMNT_PTR = INPUT_FORCMNT.FORCMNT_PTR

Uniform (Input Export Options)

Export user specified uniform load information defined in the input file.

Microsoft Access Table Name: INPUT_UNIFORM
Microsoft Excel Section Name: UNIFORM LOAD DATA
XML Primary Tag Name: UNIFORMLOAD

Data Name	Column Name	XML Tag Name
Uniform Load Pointer	UNIF_PTR	
Uniform Load Number	VECTOR_NUM	VECTOR_NUM
X axis load	UX	UX
Y axis load	UY	UY
Z axis load	UZ	UZ
Uniform Load Type	UTYPE	

Access and Excel contain a number that identifies on which element they were defined.

INPUT_BASIC_ELEMENT_DATA.ULOAD_PTR = INPUT_UNIFORM.ULOAD_PTR

Wind/Wave (Input Export Options)

Export user specified wind or wave information defined in the input file.

Microsoft Access Table Name: INPUT_WIND
Microsoft Excel Section Name: WIND/WAVE DATA
XML Primary Tag Name: WIND or WAVE

Data Name	Column Name	XML Tag Name
Wind Pointer	WIND_PTR	
Entry Type	ENTRY_TYPE	Not Applicable
Wind Shape Factor OR Wave Drag Coefficient	WSHAP_WDRC	WSHAP_WDRC
Wave added mass coefficient	WADD_MASS	WADD_MASS
Wave Lift Coefficient	WLIFT_COEFF	WLIFT_COEFF
Wave Marine Growth	WMAR_GROWTH	WMAR_GROWTH
Wave Growth Density	WMAR_GROWTH _DENSITY	

Access and Excel contain a number that identifies on which element they were defined.

INPUT_BASIC_ELEMENT_DATA.WLOAD_PTR = INPUT_WIND.WIND_PTR

Offsets (Input Export Options)

Export user specified offset information defined in the input file.

Microsoft Access Table Name: INPUT_OFFSETS
Microsoft Excel Section Name: OFFSET DATA
Microsoft XML Primary Tag Name: OFFSET

Data Name	Column Name	XML Tag Name
Offsets Pointer	OFFSETS_PTR	
From node offset in X direction	FROMX	FROMX
From node offset in Y direction	FROMY	FROMY
From node offset in Z direction	FROMZ	FROMZ
To node offset in X direction	TOX	TOX
To node offset in Y direction	TOY	TOY

Data Name	Column Name	XML Tag Name
To node offset in Z direction	TOZ	TOZ

Access and Excel contain a number that identifies on which element they were defined.

INPUT_BASIC_ELEMENT_DATA.OFFSETS_PTR = INPUT_OFFSETS.OFFSETS_PTR

Allowables (Input Export Options)

Export user specified allowable stress information defined in the input file.

Microsoft Access Table Name: INPUT_ALLOWBLES
Microsoft Excel Section Name: ALLOWABLE STRESS DATA
XML Primary Tag Name: ALLOWABLESTRESS

Data Name	Column Name	XML Tag Name
Allowables Pointer	ALLOWBL_PTR	
Allowable Stress Indicator	ALLOWBL_STRESS _INDICATOR Text file: ALLOWBL_STRESS_INDI	ALLOWBL_STRESS _INDICATOR
Case Number	CASE_NUM	CASE_NUM
Cold Allowable Stress	COLD_ALLOW	COLD_ALLOW
Hot Allowable Stress	HOT_ALLOW	HOT_ALLOW*
Cyclic Reduction Factor	CYC_RED_FACTOR	CYC_RED_FACTOR*
Design Factor	DESIGN_FACTOR	DESIGN_FACTOR
Eff	EFF	EFF
Sy (yield stress)	SY	SY
Hoop Stress	HOOP_STRESS	HOOP_STRESS
Hot Sy	HOT_SY	
SU (minimum ultimate tensile strength)	SU	
Hot SU	HOT_SU	
FAC	FAC	FAC

Data Name	Column Name	XML Tag Name
PMax	PMAX	PMAX
Code (page 225)	PIPING_CODE	PIPING_CODE
Cycles for BW (butt-weld) fatigue pair	BUTTWELDCYCLES	BUTTWELDCYCLES*
Stress for BW fatigue pair	BUTTWELDSTRESS	BUTTWELDSTRESS*
Cycles for FW (fillet-weld) fatigue pair	FILLETWELDCYCLES	FILLETWELDCYCLES*
Stress for FW fatigue pair	FILLETWELDSTRESS	FILLETWELDSTRESS*
Appendix P - Operating Allowable Reduction	APP_P_OPE_ALL _REDUCTION	
* These tags are child tags of CASE_NUM (can range from 1 through 9)		

Access and Excel contain a number that identifies on which element they were defined.

INPUT_BASIC_ELEMENT_DATA.ALLOW_PTR = INPUT_ALLOWBLS.ALLOW_PTR

SIFs & Tees (Input Export Options)

Export user specified SIF information defined in the input file.

Microsoft Access Table Name: INPUT_SIFTEES
Microsoft Excel Section Name: SIF DATA
XML Primary Tag Name: SIF

Data Name	Column Name	XML Tag Name
SIF Pointer	SIF_PTR	
SIF Number	SIF_NUM	SIF_NUM
Node Number	NODE	NODE
Type	TYPE	TYPE
SIF In-Plane	SIF_IN	SIF_IN
SIF Out-Plane	SIF_OUT	SIF_OUT
SIF Torsion	SIF_TORSION	SIF_TORSION
SIF Axial	SIF_AXIAL	SIF_AXIAL

Data Name	Column Name	XML Tag Name
SIF Pressure	SIF_PRESSURE	SIF_PRESSURE
B31.3 2010 Stress Index for in-plane, Iin	STRESSINDEX_Iin	STRESSINDEX_Iin
B31.3 2010 Stress Index for out-plane, Iout	STRESSINDEX_Iout	STRESSINDEX_Iout
B31.3 2010 Stress Index for torsion stress, It	STRESSINDEX_It	STRESSINDEX_It
B31.3 2010 Stress Index for axial stress, Ia	STRESSINDEX_Ia	STRESSINDEX_Ia
B31.3 2010 Stress Index for pressure, Ipr	STRESSINDEX_Ipr	STRESSINDEX_Ipr
Circumferential Weld	WELD_d	WELD_D
Fillet	FILLET	FILLET
Pad Thickness	PAD_THK	PAD_THK
FTG Ro	FTG_RO	FTG_RO
Crotch	CROTCH	CROTCH
Weld ID	WELD_ID	WELD_ID
B1	B1	B1
B2	B2	B2

Access and Excel contain a number that identifies on which element they were defined.

INPUT_BASIC_ELEMENT_DATA.INT_PTR = INPUT_SIFTEES.SIF_PTR

Flexible Nozzles (Input Export Options)

Export user specified nozzle information defined in the input file.

WRC 297

Microsoft Access Table Name: INPUT_NOZZLES_WRC297
Microsoft Excel Section Name: WRC297 NOZZLE DATA
XML Primary Tag Name: WRC297_NOZZLE

Data Name	Column Name	XML Tag Name
Nozzle Pointer	NOZ_PTR	
Vessel Node Number	VESSEL_NODE	VESSEL_NODE
Nozzle Node Number	NOZZLE_NODE	NOZZLE_NODE
Nozzle Outside Diameter	NOZ_OD	NOZ_OD
Nozzle Wall Thickness	NOZ_WT	NOZ_WT
Vessel Outside Diameter	VES_OD	VES_OD
Vessel Wall Thickness	VES_WT	VES_WT
Vessel Reinforcing Pad Thickness	VES_RPT	VES_RPT
Dist. to stiffeners or head	DIST_HEAD	DIST_HEAD
Dist. to opposite side stiffeners or head	DIST_OPP_HEAD	DIST_OPP_HEAD
Vessel centerline direction vector X	VES_CENT_X	VES_CENT_X
Vessel centerline direction vector Y	VES_CENT_Y	VES_CENT_Y
Vessel centerline direction vector Z	VES_CENT_Z	VES_CENT_Z
Vessel Temperature	VES_TEMP	VES_TEMP
Vessel Material #	VES_MAT	VES_MAT
Material Name	MATERIAL_NAME	MATERIAL_NAME

Access and Excel contain a number that identifies on which element they were defined.

INPUT_BASIC_ELEMENT_DATA.NOZ_PTR = INPUT_NOZZLES_WRC297.NOZ_PTR

API 650

Microsoft Access Table Name: INPUT_NOZZLES_API650
Microsoft Excel Section Name: API650 NOZZLE DATA
XML Primary Tag Name: API650_NOZZLE

Data Name	Column Name	XML Tag Name
Nozzle Pointer	NOZ_PTR	
Tank Node Number	TANK_NODE	TANK_NODE
Nozzle Node Number	NOZZLE_NODE	NOZZLE_NODE
Nozzle Outside Diameter	NOZ_OD	NOZ_OD
Nozzle Wall Thickness	NOZ_WT	NOZ_WT
Tank Outside Diameter	TANK_OD	TANK_OD
Tank Wall Thickness	TANK_WT	TANK_WT
Reinforcing	REINFORCE	REINFORCE
Nozzle height	NOZ_HEIGHT	NOZ_HEIGHT
Fluid height	FLUID_HEIGHT	FLUID_HEIGHT
Fluid specific gravity	FLUID_SG	FLUID_SG
Thermal expansion coefficient	THERM_EXP_COEFF	THERM_EXP_COEFF
Temperature change	DELTAT	DELTAT
Elastic modulus	EMOD	EMOD

Access and Excel contain a number that identifies on which element they were defined.

INPUT_BASIC_ELEMENT_DATA.NOZ_PTR = INPUT_NOZZLES_API650.NOZ_PTR

PD 5500

Microsoft Access Table Name: INPUT_NOZZLES_BS5500
Microsoft Excel Section Name: BS5500 NOZZLE DATA
XML Primary Tag Name: BS5500_NOZZLE

Data Name	Column Name	XML Tag Name
Nozzle Pointer	NOZ_PTR	
Vessel Node Number	VESSEL_NODE	VESSEL_NODE

Data Name	Column Name	XML Tag Name
Nozzle Node Number	NOZZLE_NODE	NOZZLE_NODE
Vessel Type	VESSEL_TYPE	VESSEL_TYPE
Nozzle Outside Diameter	NOZ_OD	NOZ_OD
Vessel Outside Diameter	VES_OD	VES_OD
Vessel Wall Thickness	VES_WT	VES_WT
Vessel Reinforcing Pad Thickness	VES_RPT	VES_RPT
Dist. to stiffeners or head	DIST_HEAD	DIST_HEAD
Dist. to opposite side stiffeners or head	DIST_OPP_HEAD	DIST_OPP_HEAD
Vessel centerline direction vector X	VES_CENT_X	VES_CENT_X
Vessel centerline direction vector Y	VES_CENT_Y	VES_CENT_Y
Vessel centerline direction vector Z	VES_CENT_Z	VES_CENT_Z
Vessel Temperature	VES_TEMP	VES_TEMP
Vessel Material #	VES_MAT	VES_MAT
Material Name	MATERIAL_NAME	MATERIAL_NAME

Access and Excel contain a number that identifies on which element they were defined.

INPUT_BASIC_ELEMENT_DATA.NOZ_PTR = INPUT_NOZZLES_BS5500.NOZ_PTR

Custom

Microsoft Access Table Name: INPUT_NOZZLES_USERDEFINED
Microsoft Excel Section Name: USERDEFINED NOZZLE DATA
XML Primary Tag Name: USERDEFINED_NOZZLE

Data Name	Column Name	XML Tag Name
Nozzle Pointer	NOZ_PTR	
Nozzle Node Number	NOZZLE_NODE	NOZZLE_NODE
Vessel Node Number	VESSEL_NODE	VESSEL_NODE
Vessel centerline direction vector X	VES_CENT_X	VES_CENT_X

Data Name	Column Name	XML Tag Name
Vessel centerline direction vector Y	VES_CENT_Y	VES_CENT_Y
Vessel centerline direction vector Z	VES_CENT_Z	VES_CENT_Z
Axial (Nozzle)	AXIAL_TRANSLATION _STIFFNESS	AXIAL_TRANSLATION _STIFFNESS
In-Plane Bending	IN_PLANE_BENDING _STIFFNESS	IN_PLANE_BENDING _STIFFNESS
Out-of-Plane Bending	OUT_PLANE_BENDING _STIFFNESS	OUT_PLANE_BENDING _STIFFNESS
Torsional (Nozzle)	TORSIONAL_BENDING _STIFFNESS	TORSIONAL_BENDING _STIFFNESS
Nozzle Outside Diameter	NOZ_OD	NOZ_OD
Nozzle Wall Thickness	NOZ_WT	NOZ_WT
Vessel Outside Diameter	VES_OD	VES_OD
Vessel Wall Thickness	VES_WT	VES_WT
Vessel Reinforcing Pad Thickness	VES_RPT	VES_RPT
Vessel Material #	VES_MAT	VES_MAT
Material Name	USER_DEFINED_MATERIAL _NAME	USER_DEFINED_MATERIAL _NAME

Access and Excel contain a number that identifies on which element they were defined.

INPUT_BASIC_ELEMENT_DATA.NOZ_PTR = INPUT_NOZZLES_USERDEFINED.NOZ_PTR

Equipment (Input Export Options)

Export user specified equipment information defined in the input file.

Microsoft Access Table Name: INPUT_EQUIPMENT
Microsoft Excel Section Name:
XML Primary Tag Name:

Data Name	Column Name	XML Tag Name
Equipment Pointer	EQP_PTR	
	NODE1	NODE1
	FX1	FX1
	FY1	FY1
	FZ1	FZ1
	MX1	MX1
	MY1	MY1
	MZ1	MZ1
	COSX1	COSX1
	COSY1	COSY1
	COSZ1	COSZ1
	RATING1	RATING1
	METHOD1	METHOD1
	NODE2	NODE2
	FX2	FX2
	FY2	FY2
	FZ2	FZ2
	MX2	MX2
	MY2	MY2
	MZ2	MZ2

Data Name	Column Name	XML Tag Name
	COSX2	COSX2
	COSY2	COSY2
	COSZ2	COSZ2
	RATING2	RATING2
	METHOD2	METHOD2

Access and Excel contain a number that identifies on which element they were defined.

INPUT_BASIC_ELEMENT_DATA.EQP_PTR = INPUT_EQUIPMENT.EQP_PTR

Title (Input Export Options)

Export user specified title information defined in the input file.

Microsoft Access Table Name: INPUT_TITLE
Microsoft Excel Section Name:
XML Primary Tag Name:

Data Name	Column Name	XML Tag Name
Title	TITLE	TITLE

Hangers (Input Export Options)

Export user specified hanger information defined in the input file.

Microsoft Access Table Name: INPUT_HANGERS
Microsoft Excel Section Name: HANGER DATA
XML Primary Tag Name: HANGER

Data Name	Column Name	XML Tag Name
Hanger pointer	HANGER_PTR	
Node number	NODE	NODE
Connecting node	CNODE	CNODE
Constant effort support load or stiffness	CONST_EFF_LOAD	CONST_EFF_LOAD
Load variation	LOAD_VAR	LOAD_VAR

Data Name	Column Name	XML Tag Name
Rigid support displacement criteria	RIGID_SUP	RIGID_SUP
Available space for hanger	AVAIL_SPACE	AVAIL_SPACE
Theoretical cold load	COLD_LOAD	COLD_LOAD
Operating load	HOT_LOAD	HOT_LOAD
Maximum travel limit	MAX_TRAVEL	MAX_TRAVEL
Multiple load case option	MULTI_LC	MULTI_LC
Anchor to be freed #1	FREEANCHOR1	FREEANCHOR1
Anchor to be freed #2	FREEANCHOR2	FREEANCHOR2
Degree of freedom for #1	DOFTYPE1	DOFTYPE1
Degree of freedom for #2	DOFTYPE2	DOFTYPE2
Number of hangers	NUM_HGR	NUM_HGR
Hanger table	HGR_TABLE	HGR_TABLE
Short range springs	SHORT_RANGE	SHORT_RANGE
Tag	HGR_TAG	TAG
GUID	HGR_GUID	GUID

Access and Excel contain a number that identifies on which element they were defined.

INPUT_BASIC_ELEMENT_DATA.HGR_PTR = INPUT_HANGERS.HGR_PTR

Reducers (Input Export Options)

Export user specified reducer information defined in the input file.

Microsoft Access Table Name: INPUT_REDUCERS
Microsoft Excel Section Name:
XML Primary Tag Name:

Data Name	Column Name	XML Tag Name
Reducer pointer	RED_PTR	
"To" node diameter	DIAMETER2	DIAMETERS2

Data Name	Column Name	XML Tag Name
"To" node thickness	THICKNESS2	THICKNESS2
Reducer transition slope	ALPHA	ALPHA
Reducer large end transition radius	R1	R1
Reducer small end transition radius	R2	R2

Access and Excel contain a number that identifies on which element they were defined.

INPUT_BASIC_ELEMENT_DATA.RED_PTR = INPUT_REDUCERS.RED_PTR

Flanges (Input Export Options)

Export user specified flange information defined in the input file.

Microsoft Access Table Name: INPUT_FLANGES
Microsoft Excel Section Name:
XML Primary Tag Name:

Data Name	Column Name	XML Tag Name
Flange Pointer	FLANGE_PTR	
Flange Location	FLANGE_LOCATION	FLANGE_LOCATION
Flange Evaluation Method	METHOD	METHOD
Flange Class and Material Grade	CLASS_GRADE	CLASS_GRADE
Gasket Diameter	GASKET_DIAMETER	GASKET_DIAMETER
Bold Circle Diameter	BOLT_CIRCLE_DIA	BOLT_CIRCLE_DIA
Bold Area	BOLT_AREA	BOLT_AREA
Flange Yield Stress - Cold (Ambient) Temperature	SY_COLD	SY_COLD
Flange Yield Stress, Operating Temperature 1	SY1	SY1

Data Name	Column Name	XML Tag Name
Flange Yield Stress, Operating Temperature 2	SY2	SY2
Flange Yield Stress, Operating Temperature 3	SY3	SY3
Flange Yield Stress, Operating Temperature 4	SY4	SY4
Flange Yield Stress, Operating Temperature 5	SY5	SY5
Flange Yield Stress, Operating Temperature 6	SY6	SY6
Flange Yield Stress, Operating Temperature 7	SY7	SY7
Flange Yield Stress, Operating Temperature 8	SY8	SY8
Flange Yield Stress, Operating Temperature 9	SY9	SY9
Flange Temperature	TEMPERATURE1 through TEMPERATURE24	TEMPERATURE1 through TEMPERATURE24
Flange Pressure	PRESSURE1 through PRESSURE24	PRESSURE1 through PRESSURE24

Access and Excel contain a number that identifies on which element they were defined.

INPUT_BASIC_ELEMENT_DATA.FLG_PTR = INPUT_FLANGES.FLG_PTR